CBSE Class 12 Maths — Half-Yearly Predicted Paper (Set 8 of 10)

This is Set 8 of the highly targeted Half-Yearly Predicted Paper series for CBSE Class 12 Maths (2026-27). This paper covers the ~80% half-yearly syllabus comprehensively. Topics like Linear Programming and Probability have been completely excluded as per your instructions. Every question in this set is entirely brand new and crafted based on frequent PYQ trends.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. Find the range of the function $f(x) = \frac{|x-2|}{x-2}$, where $x \neq 2$. * **Chapter:** Relations and Functions * **Source:** [PYQ 2019] [Confidence: Very High]

Q2. Find the domain of the inverse trigonometric function $f(x) = \sin^{-1}(\sqrt{x-1})$. * **Chapter:** Inverse Trigonometric Functions * **Source:** [Practice] [Confidence: High]

Q3. If A and B are symmetric matrices of the same order, then what type of matrix is $AB - BA$? * **Chapter:** Matrices * **Source:** [PYQ 2020] [Confidence: Very High]

Q4. If A is a square matrix of order 3×3 such that $A(\text{adj}A) = 10I$, find the exact value of $|\text{adj}A|$. * **Chapter:** Determinants * **Source:** [PYQ 2022] [Confidence: Very High]

Q5. What is the derivative of $\cos(x^3)$ with respect to x? * **Chapter:** Continuity and Differentiability * **Source:** [Practice] [Confidence: Medium]

Q6. The rate of change of the area of a circle with respect to its radius r at $r = 5$ cm is: * **Chapter:** Application of Derivatives * **Source:** [PYQ 2023] [Confidence: High]

Q7. Evaluate the indefinite integral: $\int \cot^2 x \, dx$. * **Chapter:** Integrals * **Source:** [PYQ 2018] [Confidence: High]

Q8. Evaluate the definite integral: $\int_{-1}^{1} x^3 e^{x^4} \, dx$. * **Chapter:** Integrals * **Source:** [PYQ 2020] [Confidence: Very High]

Q9. Find the order of the differential equation whose general solution is given by $y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}$, where c_1, c_2, c_3 are arbitrary constants. * **Chapter:** Differential Equations * **Source:** [PYQ 2019] [Confidence: High]

Q10. Write the integrating factor (I.F.) for the linear differential equation: $x\frac{dy}{dx} + 3y = x^2$. * **Chapter:** Differential Equations * **Source:** [Practice] [Confidence: High]

Q11. If $\vec{a}$ and $\vec{b}$ are non-zero vectors such that $\vec{a} \cdot \vec{b} = \frac{1}{2}|\vec{a}||\vec{b}|$, find the angle between them. * **Chapter:** Vector Algebra * **Source:** [PYQ 2022] [Confidence: Very High]

Q12. Find the magnitude of the projection of vector $2\hat{i} - \hat{j} + \hat{k}$ on the x-axis. * **Chapter:** Vector Algebra * **Source:** [PYQ 2017] [Confidence: Medium]

Q13. Write the equation of the z-axis in symmetric Cartesian form. * **Chapter:** Three-Dimensional Geometry * **Source:** [PYQ 2019] [Confidence: High]

Q14. Find the coordinates of the mid-point of the line segment joining $(2, 3, 4)$ and $(4, 1, -2)$. * **Chapter:** Three-Dimensional Geometry * **Source:** [Practice] [Confidence: High]

Q15. Let $A = \{1, 2, 3\}$. Is the relation $R = \{(1, 1), (2, 2)\}$ an identity relation? (Yes/No) * **Chapter:** Relations and Functions * **Source:** [Practice] [Confidence: Medium]

Q16. If $A = \begin{pmatrix} \alpha & 0 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$, find the real value(s) of α for which $A^2 = B$. * **Chapter:** Matrices * **Source:** [PYQ 2018] [Confidence: Very High]

Q17. What is the equation of the normal to the curve $y = \sin x$ at the origin $(0, 0)$? * **Chapter:** Application of Derivatives * **Source:** [PYQ 2020] [Confidence: High]

Q18. Find the principal value of $\cos(\sec^{-1} x + \csc^{-1} x)$, where $|x| \geq 1$. * **Chapter:** Inverse Trigonometric Functions * **Source:** [PYQ 2021] [Confidence: Very High]

Q19. (Assertion-Reason): Assertion (A): The function $f(x) = \log_e x$ is strictly increasing on the interval $(0, \infty)$. **Reason (R):** For $f(x) = \log_e x$, the derivative $f'(x) > 0$ for all $x \in (0, \infty)$. Choose the correct option (A, B, C, or D). * **Chapter:** Application of Derivatives * **Source:** [Practice] [Confidence: High]

Q20. (Assertion-Reason): Assertion (A): $\int_0^{\pi/2} \sin^3 x \, dx = \int_0^{\pi/2} \cos^3 x \, dx$. **Reason (R):** For definite integrals, $\int_0^a f(x) \, dx = \int_0^a f(a - x) \, dx$. Choose the correct option (A, B, C, or D). * **Chapter:** Integrals * **Source:** [PYQ 2023] [Confidence: Very High]

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Express $\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right)$, where $-\frac{\pi}{2} < x < \frac{\pi}{2}$, in its simplest form. * **Chapter:** Inverse Trigonometric Functions * **Source:** [PYQ 2019] [Confidence: Very High]

Q22. If $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$, show that $A^2 - 5A + 7I = O$, where I is the identity matrix of order 2. * **Chapter:** Matrices * **Source:** [PYQ 2020] [Confidence: High]

Q23. Find $\frac{dy}{dx}$ if $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$. * **Chapter:** Continuity and Differentiability * **Source:** [PYQ 2018] [Confidence: Very High]

Q24. Use differentials to approximate the value of $\sqrt{36.6}$. * **Chapter:** Application of Derivatives * **Source:** [PYQ 2017] [Confidence: Medium]

Q25. Find the area of a triangle having vertices $A(1, 1, 1)$, $B(1, 2, 3)$, and $C(2, 3, 1)$ using vector methods. * **Chapter:** Vector Algebra * **Source:** [PYQ 2022] [Confidence: High]

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Show that the relation R in the set $\mathbb{Z}$ of integers given by $R = \{(a, b) : 5 \text{ divides } (a - b)\}$ is an equivalence relation. * **Chapter:** Relations and Functions * **Source:** [PYQ 2020] [Confidence: Very High]

Q27. Using properties of determinants, prove that: $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a - b)(b - c)(c - a)$. * **Chapter:** Determinants * **Source:** [PYQ 2019] [Confidence: Very High]

Q28. Differentiate the function $y = x^{\sin x}$ with respect to x using logarithmic differentiation. * **Chapter:** Continuity and Differentiability * **Source:** [PYQ 2021] [Confidence: High]

Q29. Evaluate the indefinite integral using the half-angle substitution method: $\int \frac{dx}{5+4\cos x}$. * **Chapter:** Integrals * **Source:** [PYQ 2018] [Confidence: High]

Q30. Solve the linear differential equation: $x \log x \frac{dy}{dx} + y = \frac{2}{x} \log x$. * **Chapter:** Differential Equations * **Source:** [PYQ 2020] [Confidence: Very High]

Q31. Find the coordinates of the exact point where the straight line passing through the points $(3, -4, -5)$ and $(2, -3, 1)$ crosses the plane given by $2x + y + z = 7$. * **Chapter:** Three-Dimensional Geometry * **Source:** [PYQ 2019] [Confidence: High]

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Use the matrix inverse method to solve the following system of linear equations: $x + y + z = 6$ $y + 3z = 11$ $x - 2y + z = 0$ * **Chapter:** Matrices and Determinants * **Source:** [Practice] [Confidence: Very High]

Q33. Show mathematically that the right circular cone of least curved surface area and a given constant volume has an altitude equal to $\sqrt{2}$ times the radius of its base. * **Chapter:** Application of Derivatives * **Source:** [PYQ 2019] [Confidence: Very High]

Q34. Evaluate the following definite integral using limit properties: $\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} \, dx$. * **Chapter:** Integrals * **Source:** [PYQ 2023] [Confidence: Very High]

Q35. Using the method of integration, sketch the region and find the exact area bounded by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$. * **Chapter:** Application of Integrals * **Source:** [PYQ 2020] [Confidence: High]

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Matrices (Factory Supply & Profit) A manufacturing company produces three products X, Y, and Z which it sells in two different markets. The annual sales volumes are given below: Market I sells 10,000 units of X, 2,000 units of Y, and 18,000 units of Z. Market II sells 6,000 units of X, 20,000 units of Y, and 8,000 units of Z. The unit sale prices of X, Y, and Z are ₹2.50, ₹1.50, and ₹1.00 respectively. The unit manufacturing costs of X, Y, and Z are ₹2.00, ₹1.00, and ₹0.50 respectively. *(i)* Represent the annual sales using a 2×3 matrix and unit sale prices using a column matrix. *(1 Mark) (ii)* Using matrix multiplication, find the total revenue collected in Market I and Market II. *(1 Mark) (iii)* Find the total gross profit made by the company in both markets combined. *(2 Marks)* * **Chapter:** Matrices * **Source:** [PYQ 2018] [Confidence: High]

Q37. Case Study 2: Differential Equations (Bank Compound Interest) In a national bank, the principal deposited in an account increases continuously at the rate of $r\%$ per year. A customer deposited ₹100 into the bank. The bank advertises that this principal amount will double itself in exactly 10 years. (Given $\log_e 2 = 0.6931$) *(i)* Formulate the first-order differential equation relating the principal P with respect to time t. *(1 Mark) (ii)* Solve the differential equation to find the general solution for P at any time t. *(1 Mark) (iii)* Calculate the exact rate of interest r to verify the bank's advertisement. *(2 Marks)* * **Chapter:** Differential Equations * **Source:** [PYQ 2022] [Confidence: Very High]

Q38. Case Study 3: 3D Geometry (Laser Beam Optics) During a physics optics experiment in 3D coordinate space, a laser beam emitted from a source situated at the point $S(1, 2, 3)$ is directed along a straight line given by the Cartesian equations $\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z-3}{2}$. The laser beam hits a flat optical mirror which lies entirely on the 3D plane defined by $2x + y - z = 5$.

(i) Write the direction ratios of the laser beam's path and the normal to the optical mirror. *(1 Mark)* *(ii)* Find the exact 3D coordinates of the point where the laser beam hits the mirror. *(1 Mark)* *(iii)* Calculate the angle at which the laser beam strikes the mirror (i.e., the angle between the laser line and the plane). *(2 Marks)* * **Chapter:** Three-Dimensional Geometry * **Source:** [Practice] [Confidence: High]

Answer Key

Q1. $\{-1, 1\}$. (For $x > 2$, it is 1; for $x < 2$, it is -1). **Q2.** We require $0 \le x - 1 \le 1 \implies 1 \le x \le 2$. Domain is $[1, 2]$. **Q3.** Skew-symmetric matrix. ($(AB - BA)^T = B^T A^T - A^T B^T = BA - AB = -(AB - BA)$). **Q4.** We know $A(\text{adj}A) = |A|I$. Given it equals $10I$, so $|A| = 10$. Then $|\text{adj}A| = |A|^{n-1} = 10^{3-1} = 100$. **Q5.** $\frac{d}{dx}(\cos(x^3)) = -\sin(x^3) \cdot 3x^2 = -3x^2 \sin(x^3)$. **Q6.** Area $A = \pi r^2 \implies \frac{dA}{dr} = 2\pi r$. At $r = 5$, $\frac{dA}{dr} = 10\pi \text{ cm}^2/\text{cm}$. **Q7.** $\int(\csc^2 x - 1)dx = -\cot x - x + C$. **Q8.** 0. (Let $f(x) = x^3 e^{x^4}$. $f(-x) = (-x)^3 e^{(-x)^4} = -x^3 e^{x^4} = -f(x)$. Since it's an odd function, integral over $[-a, a]$ is 0). **Q9.** 3. (There are three independent arbitrary constants c_1, c_2, c_3). **Q10.** Divide by x: $\frac{dy}{dx} + \frac{3}{x}y = x$. I.F. $= e^{\int \frac{3}{x}dx} = e^{3\ln x} = x^3$. **Q11.** $\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = \frac{\frac{1}{2}|\vec{a}||\vec{b}|}{|\vec{a}||\vec{b}|} = \frac{1}{2} \implies \theta = 60°$ (or $\pi/3$). **Q12.** Direction vector of x-axis is $\hat{i}$. Projection $= \frac{\vec{a}\cdot\hat{i}}{|\hat{i}|} = \frac{2}{1} = 2$. **Q13.** $\frac{x}{0} = \frac{y}{0} = \frac{z}{1}$. **Q14.** $(\frac{2+4}{2}, \frac{3+1}{2}, \frac{4-2}{2}) = (3, 2, 1)$. **Q15.** No. (An identity relation must contain (a, a) for *all* $a \in A$. The element $(3, 3)$ is missing). **Q16.** $A^2 = \begin{pmatrix} \alpha^2 & 0 \\ \alpha + 1 & 1 \end{pmatrix}$. Equating to $B \implies \alpha^2 = 1 \implies \alpha = \pm 1$. Also $\alpha + 1 = 5 \implies \alpha = 4$. Since no single value of α satisfies both, there is no real value of α. **Q17.** $y = \sin x \implies y' = \cos x$. At $x = 0$, $m_{\text{tangent}} = 1$. Thus, $m_{\text{normal}} = -1$. Eq of normal: $y - 0 = -1(x - 0) \implies x + y = 0$. **Q18.** $\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}$. Thus, $\cos(\frac{\pi}{2}) = 0$. **Q19.** Option A. (Both A and R are true, and R is the correct explanation of A). **Q20.** Option A. (Both A and R are true, and R is the correct explanation of A).

Q21. $\frac{\cos x}{1 - \sin x} = \frac{\cos^2(x/2) - \sin^2(x/2)}{(\cos(x/2) - \sin(x/2))^2} = \frac{\cos(x/2) + \sin(x/2)}{\cos(x/2) - \sin(x/2)} = \frac{1 + \tan(x/2)}{1 - \tan(x/2)} = \tan(\frac{\pi}{4} + \frac{x}{2})$. Hence, $\tan^{-1}(\dots) = \frac{\pi}{4} + \frac{x}{2}$. **Q22.** $A^2 = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 8 & 5 \\ -5 & 3 \end{pmatrix}$. Then $A^2 - 5A + 7I = \begin{pmatrix} 8 & 5 \\ -5 & 3 \end{pmatrix} - \begin{pmatrix} 15 & 5 \\ -5 & 10 \end{pmatrix} + \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O$. **Q23.** $\frac{dx}{d\theta} = a(1 + \cos\theta)$, $\frac{dy}{d\theta} = a\sin\theta$. $\frac{dy}{dx} = \frac{a\sin\theta}{a(1 + \cos\theta)} = \frac{2\sin(\theta/2)\cos(\theta/2)}{2\cos^2(\theta/2)} = \tan(\frac{\theta}{2})$. **Q24.** Let $y = \sqrt{x}$. Take $x = 36$, $\Delta x = 0.6$. $dy = \frac{1}{2\sqrt{x}}\Delta x = \frac{1}{12} \times 0.6 = 0.05$. Approx value $= y + dy = 6 + 0.05 = 6.05$. **Q25.** $\vec{AB} = (1-1)\hat{i} + (2-1)\hat{j} + (3-1)\hat{k} = \hat{j} + 2\hat{k}$. $\vec{AC} = (2-1)\hat{i} + (3-1)\hat{j} + (1-1)\hat{k} = \hat{i} + 2\hat{j}$. Cross product $\vec{AB} \times \vec{AC} = \hat{i}(0-4) - \hat{j}(0-2) + \hat{k}(1-0) = -4\hat{i} + 2\hat{j} + \hat{k}$. Magnitude $= \sqrt{16 + 4 + 1} = \sqrt{21}$. Area $= \frac{\sqrt{21}}{2}$ sq units.

Q26. Reflexive: $a - a = 0 = 5(0)$, thus $(a, a) \in R$. Symmetric: If $a - b = 5k$, then $b - a = 5(-k)$, thus $(b, a) \in R$. Transitive: If $a - b = 5m$ and $b - c = 5n$, adding them gives $a - c = 5(m + n)$, thus $(a, c) \in R$. Hence, it is an equivalence relation. **Q27.** Apply $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$: $\begin{vmatrix} 1 & a & a^2 \\ 0 & b - a & b^2 - a^2 \\ 0 & c - a & c^2 - a^2 \end{vmatrix}$. Taking out $(b - a)$ and $(c - a)$: $(b - a)(c - a)\begin{vmatrix} 1 & b + a \\ 1 & c + a \end{vmatrix}$. Expanding: $(b - a)(c - a)[(c + a) - (b + a)] = -(a - b)(c - a)(c - b) = (a - b)(b - c)(c - a)$. **Q28.** $\log y = \sin x \log x$. Diff: $\frac{1}{y}\frac{dy}{dx} = \cos x \log x + \sin x \cdot \frac{1}{x}$. $\frac{dy}{dx} = x^{\sin x}\left(\cos x \log x + \frac{\sin x}{x}\right)$. **Q29.** Put $\tan(x/2) = t \implies dx = \frac{2dt}{1+t^2}$ and $\cos x = \frac{1-t^2}{1+t^2}$. Denominator becomes $5 + 4(\frac{1-t^2}{1+t^2}) = \frac{9+t^2}{1+t^2}$. Integral becomes $\int \frac{2dt}{t^2+3^2} = \frac{2}{3}\tan^{-1}(\frac{t}{3}) + C = \frac{2}{3}\tan^{-1}(\frac{\tan(x/2)}{3}) + C$. **Q30.** Divide by $x\log x$: $\frac{dy}{dx} + \frac{1}{x\log x}y = \frac{2}{x^2}$. IF $= e^{\int \frac{1}{x\log x}dx} = e^{\log(\log x)} = \log x$. Solution: $y\log x = \int \frac{2}{x^2}\log x \, dx$. By parts: $2[-\frac{1}{x}\log x - \int(-\frac{1}{x})\frac{1}{x}dx] = -\frac{2}{x}\log x - \frac{2}{x} + C$. **Q31.** Line through $(3, -4, -5)$ and $(2, -3, 1)$ has DRs $-1, 1, 6$. Eq: $\frac{x-3}{-1} = \frac{y+4}{1} = \frac{z+5}{6} = \lambda$. Point is $(-\lambda + 3, \lambda - 4, 6\lambda - 5)$. Crosses $2x + y + z = 7 \implies 2(-\lambda + 3) + (\lambda - 4) + (6\lambda - 5) = 7 \implies 6 - 2\lambda + \lambda - 4 + 6\lambda - 5 = 7 \implies 5\lambda - 3 = 7 \implies 5\lambda =$. Coordinate is $(1, -2, 7)$.

Q32. $|A| = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 1 & -2 & 1 \end{vmatrix} = 1(1+6) - 1(0-3) + 1(0-1) = 7 + 3 - 1 = 9 \neq 0$. Find adjoint and $A^{-1} = \frac{1}{9}\begin{pmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{pmatrix}$. $X = A^{-1}\begin{pmatrix} 6 \\ 11 \\ 0 \end{pmatrix} = \frac{1}{9}\begin{pmatrix} 42 - 33 + 0 \\ 18 + 0 - 0 \\ -6 + 33 + 0 \end{pmatrix} = \frac{1}{9}\begin{pmatrix} 9 \\ 18 \\ 27 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$. $x = 1, y = 2, z = 3$. **Q33.** Volume $V = \frac{1}{3}\pi r^2 h \implies r^2 = \frac{3V}{\pi h}$. Curved SA $S = \pi r l = \pi r\sqrt{r^2 + h^2}$. $S^2 = \pi^2 r^2(r^2 + h^2) = \pi^2(\frac{3V}{\pi h})(\frac{3V}{\pi h} + h^2) = \frac{9V^2}{h^2} + 3V\pi h$. Let $Z = S^2$. $Z'(h) = -\frac{18V^2}{h^3} + 3V\pi = 0 \implies 3V\pi = \frac{18V^2}{h^3} \implies h^3 = \frac{6V}{\pi}$. Substitute $V \implies h^3 = \frac{6}{\pi}(\frac{1}{3}\pi r^2 h) \implies h^3 = 2r^2 h \implies h^2 = 2r^2 \implies h = \sqrt{2}r$. $Z''(h) > 0$ verifies minimum. **Q34.** Let $I = \int_0^\pi \frac{x\sin x}{1+\cos^2 x}dx$. Use property: $I = \int_0^\pi \frac{(\pi - x)\sin(\pi - x)}{1+\cos^2(\pi - x)}dx = \int_0^\pi \frac{(\pi - x)\sin x}{1+\cos^2 x}dx$. Add both: $2I = \pi\int_0^\pi \frac{\sin x}{1+\cos^2 x}dx$. Let $\cos x = t \implies -\sin x dx = dt$. Limits: 1 to -1. $2I = \pi\int_{-1}^{1}\frac{dt}{1+t^2} = \pi[\tan^{-1} t]_{-1}^{1} = \pi(\pi/4 - (-\pi/4)) = \frac{\pi^2}{2}$. Thus,

$I = \frac{\pi^2}{4}$. **Q35.** Equation of upper ellipse: $y = \frac{3}{4}\sqrt{16-x^2}$. Area $= 4 \times \int_0^4 \frac{3}{4}\sqrt{16-x^2}\,dx = 3\left[\frac{x}{2}\sqrt{16-x^2} + \frac{16}{2}\sin^{-1}\left(\frac{x}{4}\right)\right]_0^4 = 3\left[0 + 8\sin^{-1}(1) - 0\right] = 3[8(\pi/2)] = 12\pi$ sq units.

Q36. *(i)* Sales matrix $S = \begin{pmatrix} 10000 & 2000 & 18000 \\ 6000 & 20000 & 8000 \end{pmatrix}$. Price matrix $P = \begin{pmatrix} 2.50 \\ 1.50 \\ 1.00 \end{pmatrix}$. *(ii)* Revenue $R = S \times P = \begin{pmatrix} 25000 + 3000 + 18000 \\ 15000 + 30000 + 8000 \end{pmatrix} = \begin{pmatrix} 46000 \\ 53000 \end{pmatrix}$. Market I Revenue = ₹46,000; Market II = ₹53,000. *(iii)* Cost matrix $C = \begin{pmatrix} 2.00 \\ 1.00 \\ 0.50 \end{pmatrix}$. Total Cost $= S \times C = \begin{pmatrix} 20000 + 2000 + 9000 \\ 12000 + 20000 + 4000 \end{pmatrix} = \begin{pmatrix} 31000 \\ 36000 \end{pmatrix}$. Gross Profit = Total Revenue - Total Cost $= (46000 + 53000) - (31000 + 36000) = 99000 - 67000 = ₹32,000$. **Q37.** *(i)* $\frac{dP}{dt} = \frac{r}{100}P$. *(ii)* Separating variables $\frac{dP}{P} = \frac{r}{100}dt \implies \ln P = \frac{rt}{100} + C \implies P = P_0 e^{rt/100}$, where P_0 is initial principal. *(iii)* $P(10) = 2P_0 \implies 2P_0 = P_0 e^{10r/100} \implies e^{r/10} = 2 \implies \frac{r}{10} = \ln 2 \implies r = 10 \times 0.6931 = 6.931\%$. **Q38.** *(i)* Laser beam DRs: $2, -1, 2$. Normal to mirror DRs: $2, 1, -1$. *(ii)* Any point on line: $Q(2\lambda+1, -\lambda+2, 2\lambda+3)$. Since it hits the plane $2x + y - z = 5$, substitute: $2(2\lambda+1) + (-\lambda+2) - (2\lambda+3) = 5 \implies 4\lambda + 2 - \lambda + 2 - 2\lambda - 3 = 5 \implies \lambda + 1 = 5 \implies \lambda = 4$. Hit point is $(9, -2, 11)$. *(iii)* Let θ be angle between line and plane. $\sin\theta = \frac{|\vec{b}\cdot\vec{n}|}{|\vec{b}||\vec{n}|} = \frac{|(2)(2)+(-1)(1)+(2)(-1)|}{\sqrt{4+1+4}\sqrt{4+1+1}} = \frac{|4-1-2|}{3\sqrt{6}} = \frac{1}{3\sqrt{6}}$. Angle $\theta = \sin^{-1}\left(\frac{1}{3\sqrt{6}}\right)$.

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.